RISHAB BALASUBRAMANIAN

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About Me

I'm a PhD student in Computer Science at Virginia Tech (advised by Dr. Tu Vu), specializing in efficient model development and alignment. My research focuses on data selection for pre-training, model interpretability, distillation and merging, and architectures for scalable and reliable AI.

Education

- **Virginia Tech** — Blacksburg, VA
  *Ph.D in Computer Science (Advisor: **Tu Vu**)* — *Aug 2024 - June 2029 (Expected)*
- **Oregon State University** — Corvallis, OR
  *M.S in Computer Science & Artificial Intelligence (Advisor: **Huazheng Wang**, **Thesis**)* — *Sep 2021 - May 2024*
- **National Institute of Technology** — India
  *B.Tech in Instrumentation And Control Engineering (**Thesis**)* — *Aug 2016 - May 2020*

Publications And Preprints

- **EMNLP 2025:** Pin-Jie Lin, *Rishab Balasubramanian*, Fengyuan Liu, Nikhil Kandpal, and Tu Vu. Efficient model development through fine-tuning transfer, 2025 **paper**
- **ICML 2024:** *Rishab Balasubramanian*, Jiawei Li, Prasad Tadepalli, Huazheng Wang, Qingyun Wu, and Haoyu Zhao. Adversarial attacks on combinatorial multi-armed bandits. In *Forty-first International Conference on Machine Learning*, 2024 **paper**
- **TMLR:** Zichen Wang, *Rishab Balasubramanian*, Hui Yuan, Chenyu Song, Mengdi Wang, and Huazheng Wang. Adversarial attacks on online learning to rank with stochastic click models. *Transactions on Machine Learning Research*, 2024 **paper**
- **NeurIPS 2024:** Zeyu Zhang, Yi Su, Hui Yuan, Yiran Wu, *Balasubramanian, Rishab*, Qingyun Wu, Huazheng Wang, and Mengdi Wang. Unified off-policy learning to rank: a reinforcement learning perspective. *Advances in Neural Information Processing Systems*, 36, 2024 **paper**
- **arXiv:** *Balasubramanian, Rishab* and Kunal Rathore. Contrastive learning for object detection. *arXiv preprint arXiv:2208.06412*, 2022 **paper**
- **IROS 2021:** *Balasubramanian, Rishab*, Lifeng Zhou, Pratap Tokekar, and PB Sujit. Risk-aware submodular optimization for stochastic travelling salesperson problem. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4720–4725. IEEE, 2021 **paper**
- **AIAA 2021:** *Balasubramanian, Rishab* and Sujit PB. A cooperative framework for autonomous landings of quadrotors using vision on a moving ugv. In *AIAA Scitech 2021 Forum*, page 1880, 2021 **paper**

Research Experience

- **Efficient Model Development Through Model Diffing** — Virginia
  *Research with **Dr. Tu Vu** (VT)* — *September 2024 - Present*
  - Worked on efficiently transferring post-training updates between model versions via model diffing without additional training.
  - Showed that model diffed weights act as stronger checkpoints than the base model and can be further fine-tuned on.
  - Currently working on transferring capabilities across model sizes.
  - Work accepted to **(EMNLP 2025)**

- **Adversarial Attacks on Multi-Armed Bandits** Oregon
  *Research with **Dr. Huazheng Wang** (OSU)* *September 2022 - Present*
  - Led the work to design a novel adversarial attack algorithm on combinatorial multi-armed bandits (CMAB)
  - Showed the theoretical difficulty in attacking CMAB environments in white-box and black-box settings, and provided the *first ever* algorithm to attack CMAB instances with sublinear cost.
  - Simultaneously, it was proved that online learning-to-rank can be formulated as an offline reinforcement learning problem.
  - Results have been accepted to **(ICML 2024)** **(Neurips 2023)** and **(TMLR)**.

- **Block-wise Trainable Neural Networks (code)** Remote
  *Research with **Dr. Beidi Chen** (Meta)* *August 2022 - December 2022*
  - Developed a method to train neural network architectures sequentially to reduce training time, memory consumption, and FLOPs.
  - Enforced gradient blocking and data pruning methods to further improve efficiency, reducing training time by $\sim 30\%$ while limiting drop in test accuracy to $< 2\%$.
  - Tested on a variety of networks including Transformers, ResNets and VGG models

- **3D Reconstruction from Endoscopy Images** Remote
  *Research Scientist at **EndovisionAI*** *January 2021 - April 2021*
  - Trained an encoder-decoder model using transformation consistency losses for unsupervised depth estimation from RGB endoscopy images.
  - Created an API for visualizing 3D pointclouds from the output depthmap.

- **Risk Averse Travelling Salesman Problem** Remote
  *Research with **Dr. Sujit** (IISER Bhopal) and **Dr. Tokekar** (University of Maryland)* *June 2020 - November 2020*
  - Developed a greedy algorithm to maximize the Conditional Value at Risk (CVaR) using a submodular function.
  - We further prove that the algorithm has a polynomial runtime lower than current SOTA and an approximation factor which is proportional to the optimal solution.
  - Finally, we also show that the suboptimality gap of the solutions is bound by a constant factor.
  - Paper accepted to **(IROS 2021)**

Teaching Experience

- TA for CS 5624: Natural Language Processing at Virginia Tech (Spring 2025)
- TA for CS 5814: Machine Learning at Virginia Tech with Dr. Chandan Reddy (Fall 2024)
- TA for AI 534: Machine Learning at Oregon State University with Dr. Xiaoli Fern (Fall 2023)
- Created material for CS 514: Algorithms at Oregon State University with Prof. Samina Ehsan (Spring 2022)
- TA for CS 325: Analysis of Algorithms at Oregon State University with Prof. Umma Reddy (Winter 2022)
- TA for ENGR 201: Electrical Fundamentals I at Oregon State University with Dr. Pallavi Dhagat (Fall 2021)
- Mentor for freshmen in the last two years of undergraduate